# Facial Landmarks Detection

*Four approaches evaluated on the MTFL dataset*
*HSV skin detection · Region growing · Viola-Jones · Full cascade pipeline*

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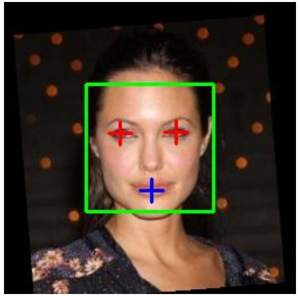
# Problem Statement & Evaluation Setup

## What we detect



- Left eye center
- Right eye center
- Mouth center

## Why classical (non-ML)

- No training data required
- Fully interpretable pipeline
- Implements core CV algorithms

## MTFL Dataset

- 12,995 face images with 5 landmark annotations
- 74% right profile — heavily non-frontal
- 136 frontal images (pose=1)

## Normalized Mean Error (NME)

```
NME = ||detected − gt|| / IOD
```

- IOD = inter-ocular distance
- Reported separately: Eye NME, Mouth NME
- Failure threshold: NME > 0.1

# Shared Pipeline Overview

| Skin / Face Detection | extractFace (morphology) | darkFeatureMask (eyes) | mouthFeatureMask (redness) | selectEyePair + selectMouth | drawLandmarks |
|---|---|---|---|---|---|
| Approach-specific | Shared | Shared | Shared | Shared | Output |

A1: HSV+YCbCr
A2: Region growing
A3: VJ face bbox
A4: VJ face bbox

Face mask → dark features → redness features → component scoring → landmarks (slides 4–6)

Landmark crosses on image

*Approaches 1–3 share everything from extractFace onward. Approach 4 replaces selectEyePair/selectMouth with cascade detectors.*
*YCbCr – luminance, chroma blue, chroma red*

# Face Extraction — extractFace()

| Opening (remove speckles) | Closing (fill holes) | Two-pass labeling | Largest CC = face | Bbox + mask + skinOnly |
|---|---|---|---|---|

**Opening (erode → dilate)**

Removes small speckles: hands, neck patches, background noise. Circular structuring element.

**Closing (dilate → erode)**

Fills small holes so face is one connected component. closingKsize < strelKsize intentionally — fills gaps without filling eye holes.

**Two-pass labeling**

As presented in laboratory. 8-connectivity forward neighborhood. BFS equivalence resolution. Assigns unique label per connected component.

**Largest connected component**

The face is the biggest skin region. All smaller components (hands, neck, hair) are discarded.

**mask vs skinOnly**

mask = post-closing (filled, used for face boundary)
skinOnly = post-opening only (eye/mouth holes preserved for feature detection)

**Bounding box**

minR, maxR, minC, maxC from single pass over labeled image. midRow, midCol used as face center for symmetry scoring.

# Eye Detection — darkFeatureMask() + selectEyePair()

## darkFeatureMask()

- Eye band: 20–55% of face height from top
- Excludes skinOnly pixels (not a dark feature)
- isInsideFace: scanline test — pixel must have face-mask pixels left AND right
- Threshold strategy:
  - Otsu (A1): maximize between-class variance
  - mean − offset (A2/A3): simpler, works when mask is tight
- Small opening after threshold to remove speckles

## selectEyePair()

**Scoring formula [Saber & Tekalp 1998]:**

```
score = |Δy|×1.0 + asym×1.0
      + (1−areaRatio)×30 − yReward×5
```

- Must straddle face vertical midline
- $|\Delta y| \leq 10\%$ of face height
- Separation $\in$ [20%, 65%] of face width
- Area ratio penalizes asymmetric blobs (×30 weight)
- yReward: prefer lower pairs (eyebrows are above eyes)
- Single-eye fallback: mirror across midline

# Mouth Detection — mouthFeatureMask() + selectMouth()

## mouthFeatureMask()

**Redness criterion [Hsu et al. 2002 — MouthMap, simplified]:**

```
R − G > 30   AND   R > B
```

- Mouth band: 68–90% (A1/A3), 80–95% (A2)
- Works on lips regardless of darkness — lips aren't always dark
- Small opening after to remove noise

## selectMouth()

- Largest redness blob scoring near midline
- Aspect filter: height ≤ 1.3 × width (mouth is wide)
- Width ∈ [15%, 90%] of face width
- Score = area × (1 − horizontal penalty)
- Y = blob.minR (top of blob, not centroid)
  - Centroid pulled below lips by chin/neck redness

*Important implementation note: the full Hsu et al. MouthMap uses $R^2/(R+G+B) \times (R-B)^2$ in YCbCr space. Currents implementation is a simplified BGR approximation.*

# Approach 1 — HSV + YCbCr Skin Detection

A1

HSV classify pixels → AND YCbCr → Skin mask → extractFace → detectLandmarks (Otsu)

### HSV Thresholds

```
H ∈ [0, 25]   S ∈ [40, 255]   V ∈ [60, 255]
```

- H separates skin tone from background (0–25 = orange-red)
- S excludes grey/white regions
- V excludes very dark pixels
- Empirically tuned [Hassan & Saud 2023, Shaik et al. 2015]

### YCbCr AND filter [Rahmat et al. 2016]

```
Cr ∈ [133, 173]   Cb ∈ [77, 127]
```

- Pixel must pass BOTH HSV AND YCbCr
- YCbCr separates luminance from chrominance differently
- Rejects warm-toned backgrounds that fool HSV
- Otsu threshold for eyes adapts to each image

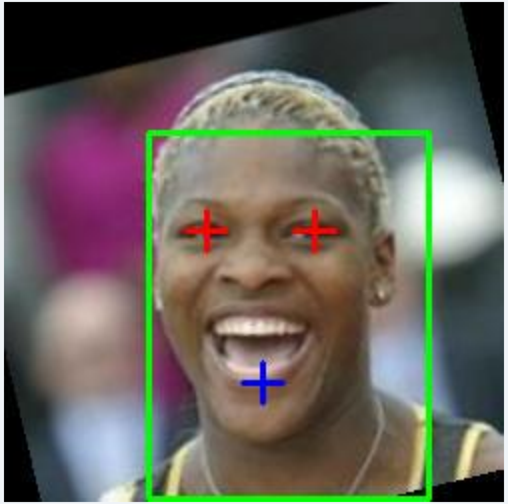
*Unique to A1: dual color space skin filter + Otsu eye threshold. Everything after skin mask is shared pipeline (slides 4–6).*

# Approach 1 — HSV + YCbCr Skin Detection

A1

*Skin mask*

*Face mask*



*Detected landmarks*

# Approach 2 — Region Growing

A2

User click (seed pixel) → HSV of seed sampled → BFS expand 4-neighbors → Skin mask → extractFace + landmarks

**Region Growing BFS [Gonzalez & Woods Ch.10]**

**Accept neighbor if:**

```
ΔH < 10   AND   ΔS < 40   AND   ΔV < 50
```

- ΔH with wraparound: if dH > 90 → dH = 180 – dH
- ΔV loose (50): forehead brighter than cheek, both skin
- Adapts to any skin tone — no fixed thresholds
- Auto-seed for evaluation: image center (img.cols/2, img.rows/2)

**A2 vs A1**

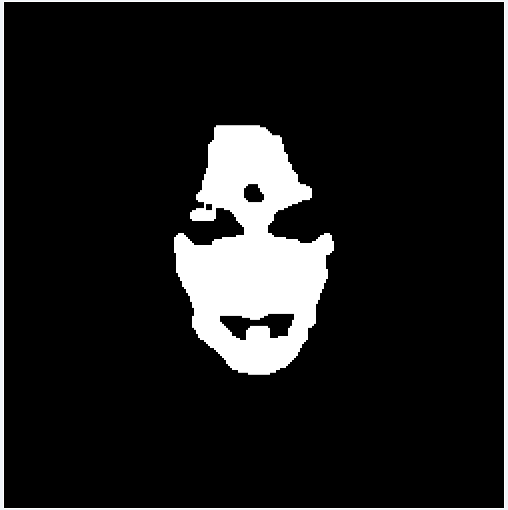
- Better NME (0.801 vs 1.121)
- Cleaner face mask → better eye hole preservation
- Lower detection rate (57.7% vs 72.1%)
- Fixed center seed misses non-centered faces
- Interactive mode (user click) performs better than auto-seed
- mean−offset threshold for eyes (not Otsu)

*Unique to A2: adaptive skin detection per image. Everything after skin mask is shared pipeline (slides 4–6).*

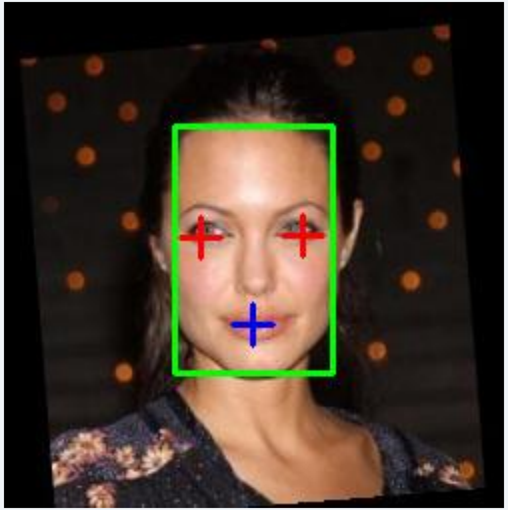
# Approach 2 — Region Growing

A2

*Skin mask*



*Face mask*



*Detected landmarks*

# GT Seed Experiment — Isolating the Bottleneck

*Hypothesis: Is the low detection rate caused by bad seed selection, or by the landmark localization algorithm?*

**Experiment: Run A2 with the ground truth eye midpoint as seed**

*(theoretical upper bound — uses GT data, not a fair evaluation)*

| Approach | Eye Det.% | Eye NME |
|---|---|---|
| A2 — center seed (fair) | 57.7% | 0.801 |
| A2 — GT seed (upper bound) | 68.7% | 0.796 |

**Finding**

- Detection rate improved: 57.7% → 68.7% (better seed = better face mask)
- **Eye NME unchanged: 0.801 → 0.796 (essentially the same)**
- Conclusion: seed quality is NOT the primary source of error
- The landmark localization algorithm is the bottleneck

# Approach 3 — Viola-Jones Face + Classical Landmarks

A3

VJ face detection → FaceGeometry (bbox, mask=rect) → detectLandmarks (offset=40)

### Viola-Jones Algorithm [Viola & Jones 2001]

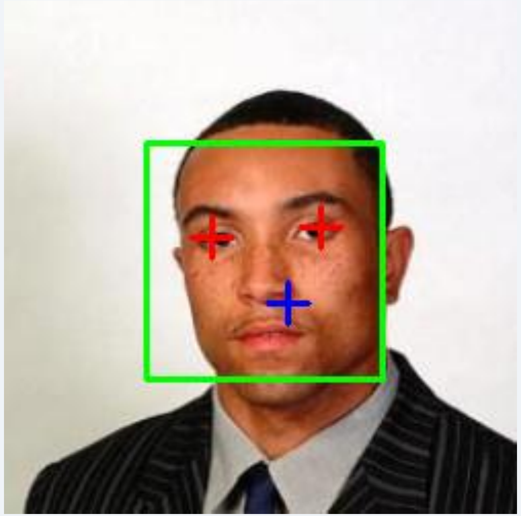
- Haar-like features: rectangular intensity difference filters
- Integral image: O(1) feature evaluation at any scale
- AdaBoost: selects most discriminative features from 160,000+ candidates
- Cascade: 38 stages, each rapidly rejects non-face windows
- Trained on thousands of face images → robust to skin tone + lighting

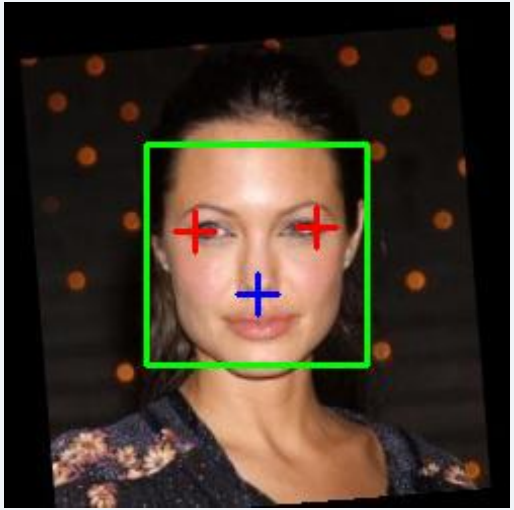
### A3 Implementation Details

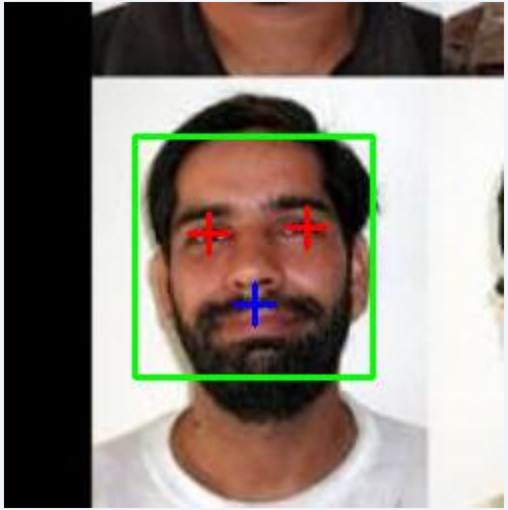
- haarcascade_frontalface_default.xml via OpenCV CascadeClassifier
- detectMultiScale: scaleFactor=1.1, minNeighbors=4, minSize=60px
- skinOnly = all zeros (no skin classification inside bbox)
  - darkFeatureMask examines all pixels inside bbox
- eyeDarknessOffset = 40 (stricter — no skin filter to help)
- mean−offset threshold (not Otsu) — bbox already tight
- **Result: 95.5% detection rate — best of all approaches**

# Approach 3 —Viola Jones Face + Classical Landmarks

A3 ***Detected landmarks***







# Approach 4 — Full Viola-Jones Pipeline

A4

VJ face detection → Eye cascade (upper ROI) → Mouth cascade (lower ROI) → Landmark points

**Eye detection — haarcascade_eye.xml**

- Search restricted to upper half of face ROI
- minNeighbors = 2 (permissive, ROI already restricted)
- Take two largest detections → left/right eye
- Sort by area, centroid → image coords

**Mouth — haarcascade_mcs_mouth.xml**

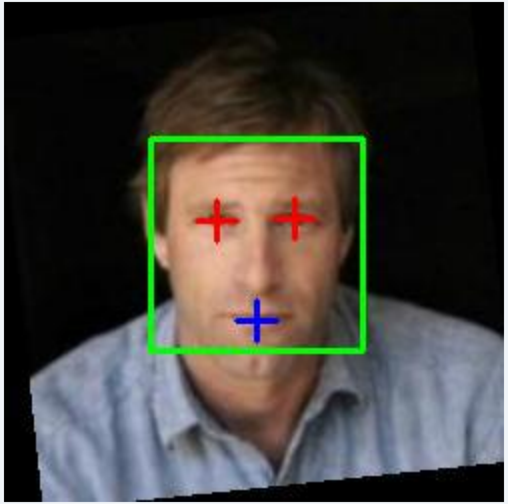
- Search restricted to lower half of face ROI
- minNeighbors = 11 (strict, mouth cascade has many false positives)
- Take largest detection
- Castrillón-Santana et al. 2007, Univ. Las Palmas, 7000+ samples

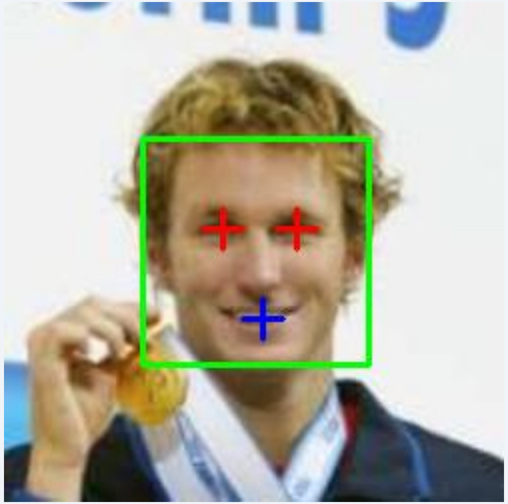
**Precision-Recall Trade-off**

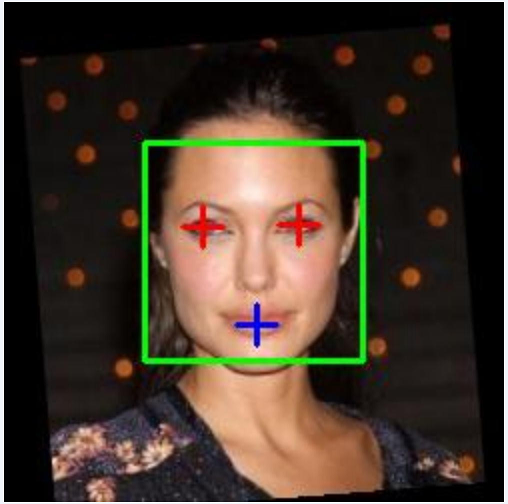
- Best NME (Eye: 0.685, Mouth: 0.403) when detected — cascades more precise than thresholding
- Lowest detection rate (39.6%) — eye cascade requires good frontal alignment

# Approach 4 — Full Viola Jones Pipeline

**A3** *Detected landmarks*

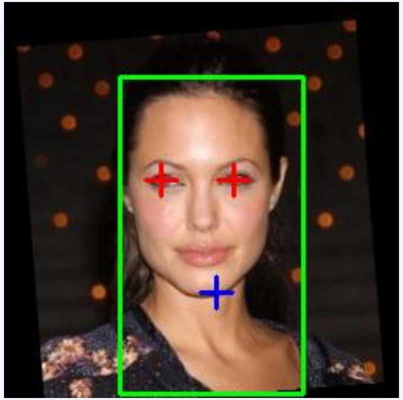




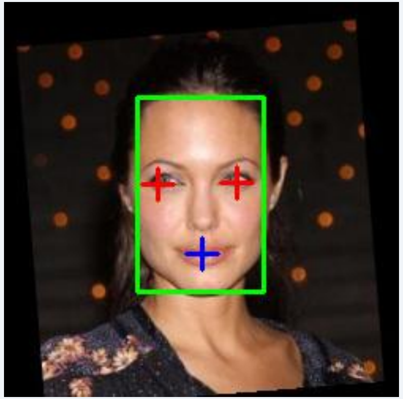


# Comparison

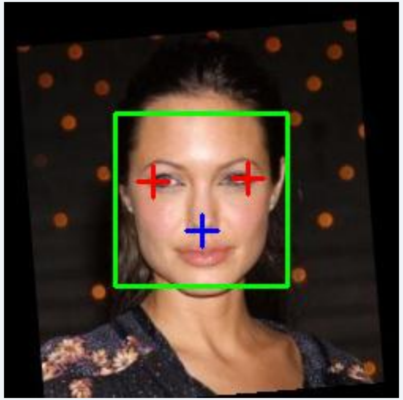
**All** *Detected landmarks*



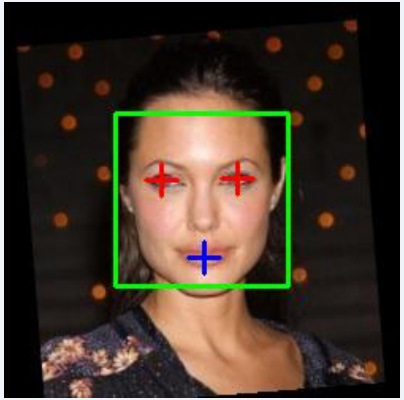
*A1*



*A2*



*A3*



*A4*

# Evaluation Framework

```
Eye NME = mean( ||leftEye_det − leftEye_gt|| / IOD,  ||rightEye_det − rightEye_gt|| / IOD )

Mouth NME = ||mouth_det − mouthCenter_gt|| / IOD    |    Failure: NME > 0.1
```

### MTFL Dataset

- 12,995 images, 5 landmark annotations each
- Evaluated on 2,000 images (training.txt)
- Separate frontal subset: 136 images (pose=1)
- IOD = distance between GT eye centers
- Mouth GT = midpoint of two mouth corners

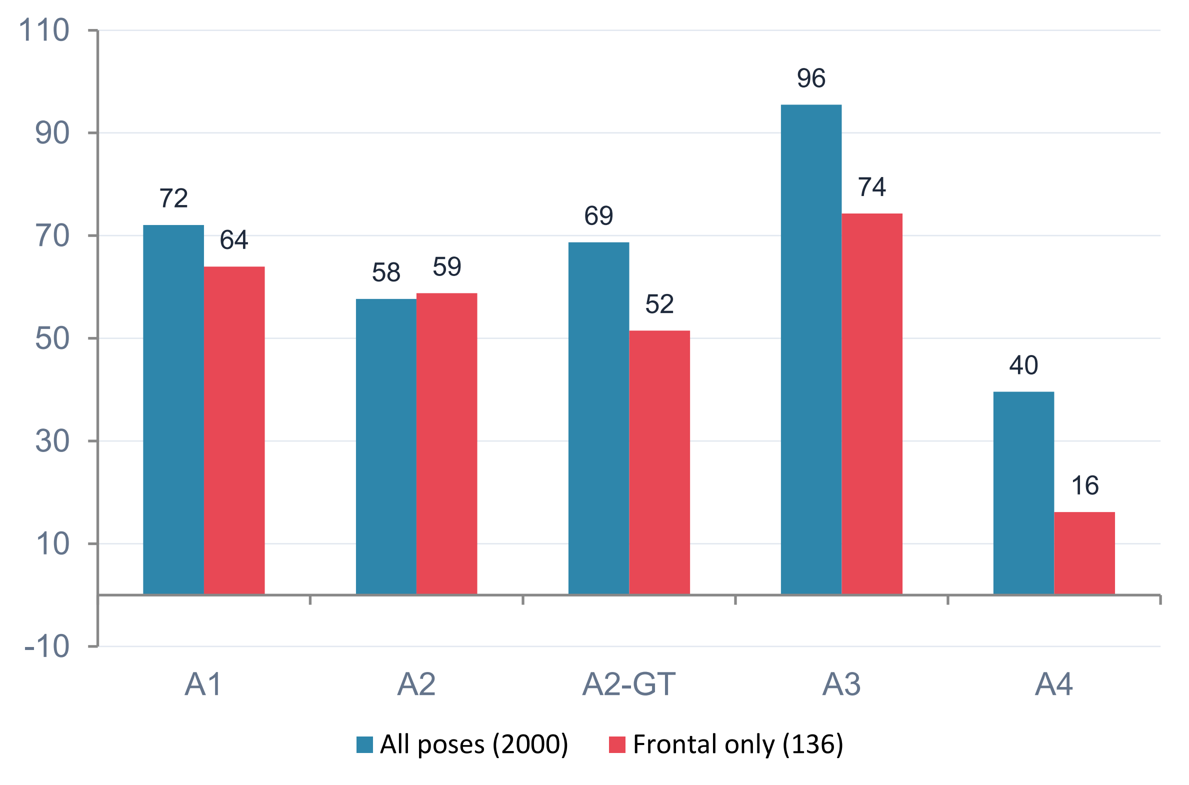
### Stratification

- pose: 1=frontal, 2=left, 3=right, 4=up, 5=down
- 74.4% right profile, only 1.4% frontal
- glasses: with (396) / without (1604)
- smile: smiling (1004) / not smiling (996)
- gender: male/female

### CSV Export

- Per-image row: image, pose, smile, glasses, gender, eye_det, mouth_det, eye_nme, mouth_nme, combined_nme
- −1 for undetected landmarks
- Python script: analyze_results.py

# Results — Eye Detection Rate




**95.5%**

A3 detection rate
— best of all

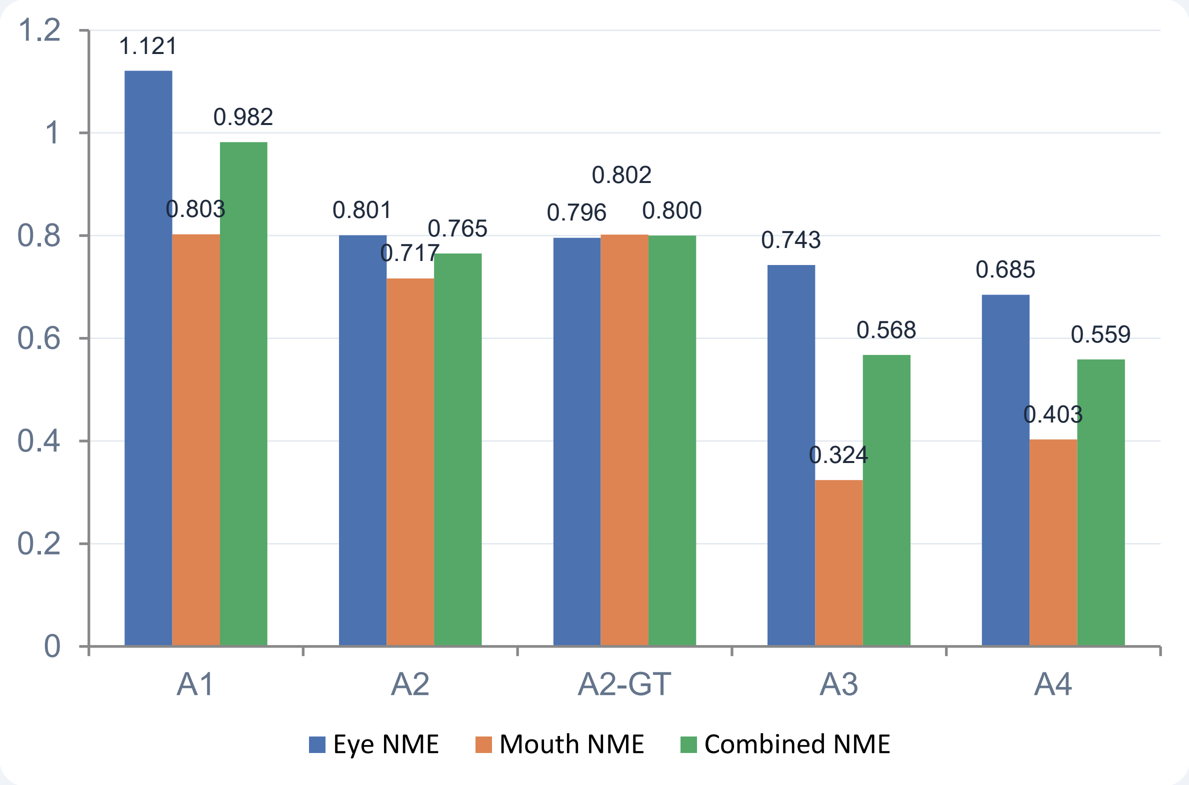
**39.6%**

A4 — eye cascade
Is very restrictive

**16.2%**

A4 frontal:
cascade collapses

# Results — NME Breakdown (All Poses, 2000 images)



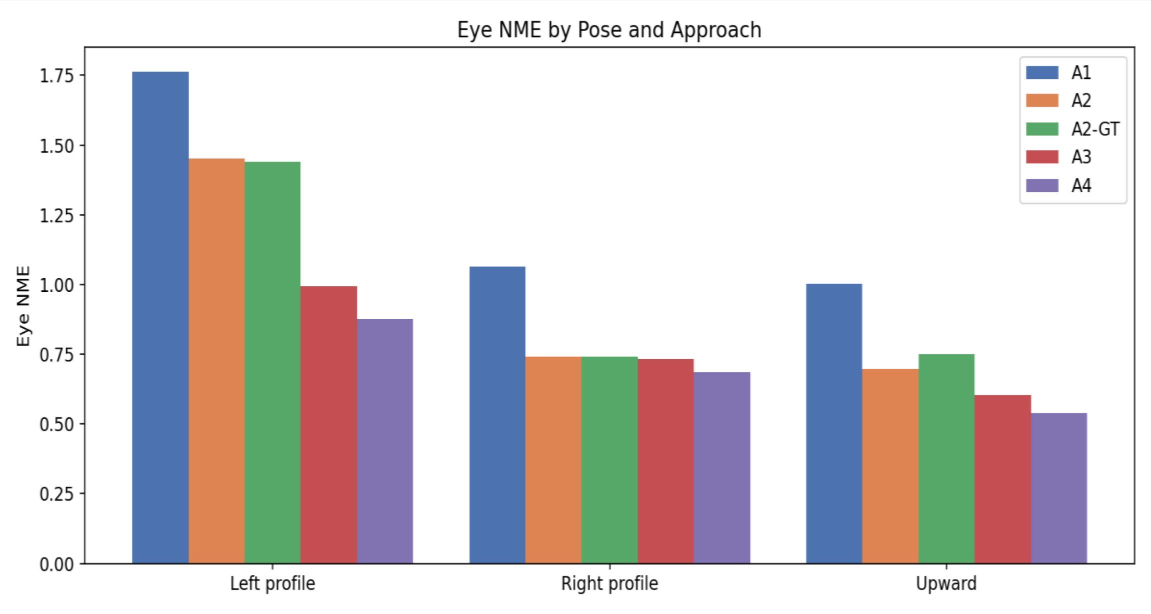

A4 best Eye NME (0.685)
Cascades more precise

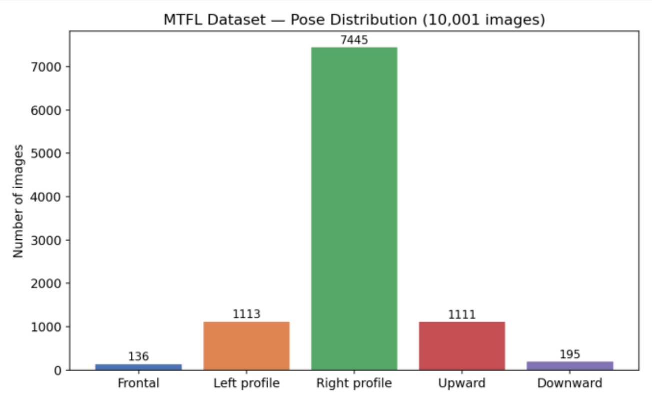
A3 best Mouth NME (0.324)
Redness works well with reliable bbox

A1 worst overall (1.121)
Noisy skin mask confuses eyes

100% failure rate all approaches
NME > 0.1 always — strict threshold

# Stratified Analysis — Pose




- Right profile (74% of dataset) — all approaches perform similarly on this dominant case
- Left profile — harder: A3/A4 still best, A1/A2 degrade significantly
- Upward tilt — easiest non-frontal: more face visible, skin detection works better
- A3/A4 most consistent across poses — VJ face detection robust to orientation

# Stratified Analysis — Frontal, Glasses, Smile

**Frontal only (136 images):**

| Approach | Eye det.% | Eye NME |
|---|---|---|
| A1 — HSV+YCbCr+Otsu | 64.0% | 2.262 |
| A2 — center seed | 58.8% | 1.940 |
| A3 — VJ+classical | 74.3% | 1.522 |
| A4 — VJ full | 16.2% | 1.091 |

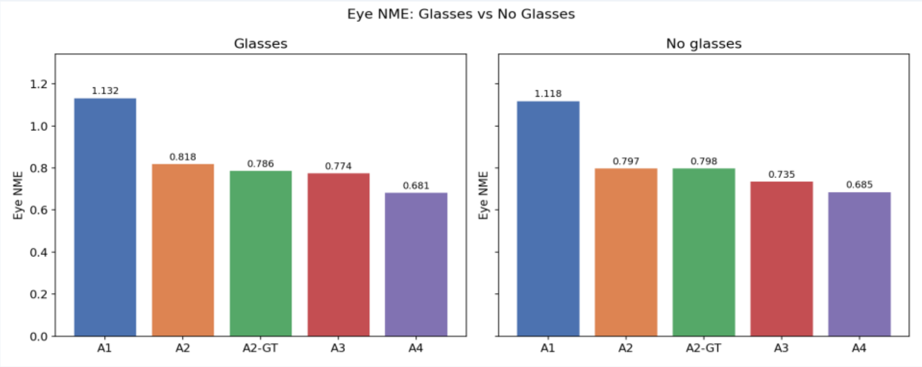
*All approaches worse on frontal than overall — MTFL frontal images are rare/atypical.*

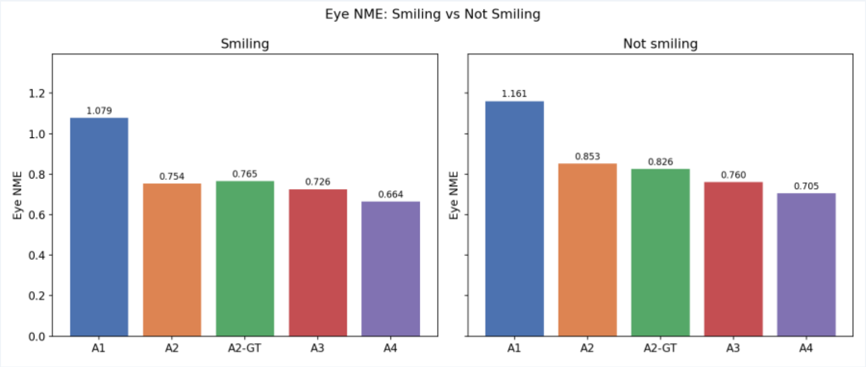
**Glasses:**

- A4: 26.5% det. with glasses vs 42.8% without — eye cascade confuses frames
- A1-A3: minimal glasses effect

**Smile:**

- Smiling slightly easier across all approaches
- Smiling portraits tend to be more frontal and well-lit



# Conclusions

01

### Landmark localization is the bottleneck — not skin detection

The GT seed experiment confirmed that even perfect skin detection (GT seed) leaves NME essentially unchanged. The symmetry-scoring algorithm and thresholding approach are the limiting factors. 100% failure rate (NME > 0.1) across all approaches.

02

### Trained models solve face detection — classical methods cannot match

A3 achieves 95.5% detection rate vs 72.1% (A1) and 57.7% (A2), using the same landmark pipeline. VJ eliminates the entire skin detection failure mode. Landmark accuracy (NME) barely changes, confirming finding #1.

03

### Precision-recall trade-off at every level

A4 achieves the best NME (Eye 0.685, Mouth 0.403) but lowest detection rate (39.6%). Glasses reduce A4 detection by 38%. Frontal images paradoxically harder — MTFL frontal images are rare and atypical. MTFL is 74% right-profile.

# Source Code and References

### Source Code and Dataset

**Source code:** https://github.com/ralu2004/Facial-Landmarks-Detection
**Dataset Source:** http://mmlab.ie.cuhk.edu.hk/projects/TCDCN/data/MTFL.zip

### References

**Cited in project README:**
https://github.com/ralu2004/Facial-Landmarks-Detection#references

### Laboratory Documentation

**Teaching Assistant website:**
https://users.utcluj.ro/~robert/ip/ip.html